

Data Science & Artificial Intelligence

Artificial Intelligence

DPP: 1

Proposition & Predictive Logic

Q1 The number of different possible logical connectives on n variables is _____.

- (A) 2^n (B) n^2
(C) 2^{2n} (D) 2^{n^2}

Q2 Consider the following statements:

S1: AI stands for Artificial Intelligence.

S2: $2 + 1 = 5$

- (A) S1 is a proposition but S2 is not a proposition
(B) S1 is not proposition but S2 is a proposition
(C) Both S1 and S2 are propositions
(D) Neither S1 nor S2 is a proposition.

Q3 The negation of the statement "We have those students who have applied for CS paper or DA paper" is

- (A) We do not have those students who have applied for CS paper or DA paper
(B) We have those students who have not applied for CS paper and not applied for DA paper
(C) We have those students who have applied for CS paper and DA paper.
(D) None of the above

Q4 "If I am selected, then I will get the job." is equivalent to saying

- (A) "If I am not selected, then I will not get the job".
(B) "If I will not get the job, then I am not selected."
(C) "If I will get the job then I am selected."
(D) None of the above

Q5 The expression $P \leftrightarrow q$ can also be stated as

- (A) "p is necessary and sufficient for q"
(B) "if p then q, and conversely"
(C) "p exactly when q."
(D) $(p \rightarrow q) \wedge (q \rightarrow p)$

Q6 Simplify the expression $(p \vee \neg q) \rightarrow (p \wedge q)$

- (A) p (B) q
(C) T (D) F

Q7 S1: $(p \vee q \vee r) \wedge (\neg p \vee \neg q \vee \neg r)$

S2: $(p \vee \neg q) \wedge (q \vee \neg r) \wedge (r \vee \neg p) \wedge (p \vee q \vee r) \wedge (\neg p \vee \neg q \vee \neg r)$

- (A) S1 is satisfiable and S2 is not satisfiable
(B) S2 is satisfiable and S1 is not satisfiable
(C) Both S1 and S2 are satisfiable
(D) Neither S1 nor S2 is satisfiable

Q8 The expression " $[(p \vee q) \wedge (p \rightarrow r) \wedge (q \rightarrow r)] \rightarrow r$ " is

- (A) Tautology
(B) Contradiction
(C) Contingency
(D) None of the above

Q9 Translate the statement: $\forall x(C(x) \vee \exists y(C(y) \wedge F(x, y)))$

into English, where $C(x)$ is "x studies Computer Science," $F(x, y)$ is "x and y are friends," and the domain

for both x and y consists of all students in your school.

- (A) Every student in your school studies Computer Science or has a friend who studies Computer Science.
(B) Every student in your school studies Computer Science or has at least one friend who studies Computer Science."
(C) There exists a student who studies Computer Science, and all students are friends with this student.
(D) Every student studies Computer Science and has at least one friend who studies Computer Science.

Q10 The number of expressions equivalent to $\forall w \neg \forall a \exists f (P(w, f) \wedge Q(f, a))$ are _____.

A: $\forall w \exists a \neg \exists f (P(w, f) \wedge Q(f, a))$

B: $\forall w \exists a \forall f \neg (P(w, f) \wedge Q(f, a))$



C: $\forall w \exists a \forall f (\neg P(w, f) \vee \neg Q(f, a))$

Q11 The premises "It is raining today, and the roads are slippery.", "We will go hiking only if it is not raining", "If we do not go hiking, then we will visit the museum." and "If we visit the museum, then we will take the bus home." lead to the conclusion

- (A) We will not go hiking and we will take the bus home
- (B) We will go hiking
- (C) we will not take the bus home
- (D) we do not visit the museum

Q12 The expression $(p \leftrightarrow q) \oplus (\neg p \leftrightarrow \neg r)$ is

- (A) Tautology
- (B) Contradiction
- (C) Contingency
- (D) None of the above

Q13 The expression $\neg(p \rightarrow q) \rightarrow \neg q$ is a.

- (A) Tautology
- (B) Contradiction
- (C) Contingency
- (D) None of the above

Q14 Determine the truth value of each of the following statements where $U = \{1, 2, 3\}$ is the universal set :

- (a) $\exists x \forall y, x^2 < y + 1$; (b) $\forall x \exists y, x^2 + y^2 < 12$;
- (A) a is True; b is False
- (B) a is False; b is True
- (C) a is True; b is True
- (D) a is False; b is False

Q15 Let $\alpha = (P \vee Q) \wedge (P \rightarrow Q) \wedge (Q \rightarrow P)$

The number of rows in the truth table having value of α as TRUE is _____.



Answer Key

Q1 (C)

Q2 (C)

Q3 (A, B)

Q4 (B)

Q5 (A, B, C, D)

Q6 (B)

Q7 (A)

Q8 (A)

Q9 (A, B)

Q10 3

Q11 (A)

Q12 (C)

Q13 (A)

Q14 (C)

Q15 1

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Hints & Solutions

Q1 Text Solution:

The number of different possible logical connectives on n variables is 2^{2^n} , because there are 2^n rows in the truth table.

Q2 Text Solution:

Both S1 and S2 are propositions as both of them are declarative statements. S1 has the truth value of TRUE whereas S2 has the truth value of FALSE.

Q3 Text Solution:

Statement A and B are equivalent using De Morgan's law

Q4 Text Solution:

p : I am selected

q : I will get the job

Given statement is $p \rightarrow$

$P \rightarrow q \equiv \sim q \rightarrow \sim P$ (Contrapositive)

Q5 Text Solution:

All are equivalent to each other

Q6 Text Solution:

The Truth table of $(p \vee \neg q) \rightarrow (p \wedge q)$				
P	q	$\neg q$	$p \vee \neg q$	$p \wedge q$
T	T	F	T	T
T	F	T	T	F
F	T	F	F	F
F	F	T	T	F

Q7 Text Solution:

$(p \vee q \vee r) \wedge (\neg p \vee \neg q \vee \neg r)$ is true when at least one of p , q , and r is true and at least one is false.

Hence, $(p \vee q \vee r) \wedge (\neg p \vee \neg q \vee \neg r)$ is satisfiable, as there is at least one assignment of truth values for p , q , and r that makes it true.

$(p \vee \neg q) \wedge (q \vee \neg r) \wedge (r \vee \neg p) \wedge (p \vee q \vee r) \wedge (\neg p \vee \neg q \vee \neg r)$ to be true, $(p \vee \neg q) \wedge (q \vee \neg r) \wedge (r \vee \neg p)$ and $(p \vee q \vee r) \wedge (\neg p \vee \neg q \vee \neg r)$

must both be true. For the first to be true, the three variables must have the same truth values, and for the second to be true, at least one of the three variables must be true and at least one must be false. However, these conditions are contradictory. From these observations we conclude that no assignment of truth values to p , q , and r makes $(p \vee \neg q) \wedge (q \vee \neg r) \wedge (r \vee \neg p) \wedge (p \vee q \vee r) \wedge (\neg p \vee \neg q \vee \neg r)$ true.

Hence, it is unsatisfiable.

Q8 Text Solution:

$$\begin{aligned}
 & [(P \vee q) \wedge (P \rightarrow r) \wedge (q \rightarrow r)] \rightarrow r \\
 &= [(P \vee q) \wedge (\neg P \vee r) \wedge (\neg q \vee r)] \rightarrow r \\
 &= [\{(p \wedge r) \vee (\neg p \wedge q) \vee (q \wedge r)\} \wedge (\neg q \vee r)] \rightarrow r \\
 &= [(p \wedge r \wedge \neg q) \vee (p \wedge r) \vee (q \wedge r)] \rightarrow r \\
 &= (p \wedge r) \rightarrow r \\
 &= \neg p \vee \neg r \vee r \\
 &= \neg p \vee T \\
 &= T.
 \end{aligned}$$

Q9 Text Solution:

A and B are equivalent to each other and they are the correct translations of the given statement

Q10 Text Solution:

All are equivalent to each other

Q11 Text Solution:

p : "It is raining today"

q : "the roads are slippery"

r : "We will go hiking"

s : "we visit the museum"

t : "we will take the bus home"

(1) $p \wedge q$ (premise)

(2) p (simplification using (1))

(3) $r \rightarrow \neg p$ (premise)

(4) $\neg r$ (Modus Tollens using 2 and 3)

(5) $\neg r \rightarrow s$ (premise)

(6) s (modus ponens using 4 and 5)

(7) $s \rightarrow t$ (premise)

(8) t (modus ponens using 6 and 7)

$\therefore \neg r \wedge t$ (Conjunction using 4 and 8).

Q12 Text Solution:


p	q	$\neg p$	$\neg r$	$p \leftrightarrow q$	$\neg p \leftrightarrow \neg q$	$(p \leftrightarrow q) \oplus (\neg p \leftrightarrow \neg r)$
T	T					
	T					
T	T					
	F					
T	F	F	F	T	T	F
	T	F	T	T	F	T
T	F	F	F	F	T	T
	F	F	T	F	F	F
F	T	T	F	F	F	F
	T	T	T	F	T	T
F	T	T	F	T	F	T
	F	T	T	T	T	F
F	F					
	T					
F	F					
	F					

Q13 Text Solution:

p	q	$p \rightarrow q$	$\neg(p \rightarrow q)$	$\neg q$	$\neg(p \rightarrow q) \rightarrow \neg q$
T	T	T	F	F	T
T	F	F	T	T	T
F	T	T	F	F	T
F	F	T	F	T	T

Q14 Text Solution:

(a) True, For if $x = 1$, then 1, 2 and 3 are all solutions to $1 < y + 1$.

(b) True. For each x_0 , let $y = 1$; then $x_0^2 + 1 < 12$ is a true statement.

Q15 Text Solution:

P	Q	$P \vee Q$	$P \rightarrow Q$	$(P \vee Q) \wedge (P \rightarrow Q)$	$Q \rightarrow P$	a
T	T	T	T	T	T	T
T	F	T	F	F	T	F
F	T	T	T	T	F	F
F	F	F	T	F	T	F



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